Technical Case Study: Day-Ahead Price Forecasting

Overview

Your task is to build a model that forecasts PJM day-ahead (DA) energy prices, then submit a forecast for a held-out period and present your approach to us.

We care about two things equally: how accurate your forecast is, and how well you reason about the problem. A simple model that is well-understood and honestly validated can beat a complex one that leaks future information.

Live forecast during the interview: as part of the interview, we will ask you to use your model to produce a DA energy price forecast for us on the spot. Make sure your code can run end to end on new input data quickly.

Data provided

We will supply the datasets below. You are free to use all, some or none of them, and to add others.

Dataset	Granularity	Notes
Load forecast, before DA	Hourly, by zone	Each row carries the time the forecast was issued
Renewables forecast (solar, wind), before DA	Hourly	Each row carries the time the forecast was issued

Natural gas prices	Daily, by hub	Key driver of marginal cost
Generation outages	Daily	Capacity offline, planned and forced
DA prices (history)	Hourly	The target variable for past periods

Why before DA matters: the DA market clears the day before delivery. Your model may only use forecasts that were available before the DA bid deadline. Using a forecast issued after that time, or using actuals in place of forecasts, is look-ahead bias and will be penalised.

The task

Forecast the hourly DA energy price (the energy component of LMP) for each hour of the test period.

- **Target:** hourly DA energy price for the RTO in $/MWh
  LMP = Energy + Congestion + Losses
- **Training period:** 2023-01-01 to 2025-12-31
- **Test period:** 2026-01-01 to 2026-08-31; we hold back the actual prices
- **Information cut-off:** each day's forecast may only use data available before 10:30 EPT on the day before delivery

Feature engineering is up to you. You can use the features we provide, select a subset, or build your own. Tell us which features you chose, which you dropped, and why.

Any modelling approach is allowed, from linear regression to gradient boosting or neural nets. Explain why your choice suits the problem.

PJM Data Miner 2

You are encouraged to pull extra data from PJM Data Miner 2, PJM's public data tool. Feeds that are often useful for DA price forecasting include generation by fuel type (gen_by_fuel), scheduled interchange, DA transmission constraints, and much more. Check the Data Availability page for the full feed list and history depth.

Getting access. Browsing data in the Data Miner 2 web interface is open to the public and needs no login (Getting Started guide). To pull data programmatically through the API, you need a PJM Tools account and a free API key, which PJM issues after you register and request Data Miner API access; see the API guide. Approval is not instant, so register early. Non-members are limited to 6 data connections per minute, and the data is for internal use only and may not be redistributed (Data Miner page). If you can't get API access in time, downloads from the web interface are fine.

As with our data, any forecast feed you add must respect the information cut-off: use the vintage that was available before the DA deadline.

Deliverables

1. **Forecast file:** a CSV with one row per hour of the test period, columns datetime_beginning_ept and da_price_forecast ($/MWh)
2. **Code:** notebooks or scripts that reproduce your forecast end to end from the raw data
3. **Presentation:** a short deck covering:
    - Data preparation and how you handled vintages and the cut-off
    - Features you built or selected, and why
    - Model choice and validation method (e.g. rolling or walk-forward backtest)

- Backtest accuracy, and where the model does well or badly (peaks, spikes, shoulder months)
- What you would do next with more time or data

4. **Live forecast:** during the interview, run your model to produce a DA energy price forecast for a period we choose, and walk us through the result

Evaluation

We evaluate you on two things: the forecast itself and the quality of your approach.

Area	What we look for
Forecast accuracy	Error on the held-out test period (MAE and RMSE in $/MWh), compared with a naive baseline such as same-hour-last-week
Data handling	Correct use of vintages, no look-ahead bias, sensible treatment of missing data and outliers
Feature engineering	Features grounded in power fundamentals, not just a large feature dump
Validation	Honest out-of-sample testing that mimics how the model would run day to day
Communication	A clear story of what drives DA prices and where the model's blind spots are

Logistics and rules

- **Time allowed:** From receipt of data until Presentation
- **Presentation length:** 10 minutes of presentation followed by 5 minutes of Q&A
- **Tools:** any language or library (Python, R, etc.)
- **Data sources:** the provided datasets, PJM Data Miner 2, and any other public source; cite everything you use
- **Questions:** email msingh@gbefund.com if anything about the data is unclear

Instructions for Requesting PJM Data Miner Access

To obtain access to Data Miner, please send an email to accountmanager@pjm.com using the template provided below:

Subject: Request for Non-Member Data Miner Access

Hello,

I am a student at [Your University Name] interested in learning more about power markets and the PJM Interconnection. Could you please grant me access to a non-member Data Miner account?

Thank you,
[Your Name]